

Revision Notes

Class – 12 Mathematics

Chapter 5 - Continuity and Differentiability

CONTINUITY

1. DEFINITION

A function $f(x)$ is said to be continuous at $x = a$; where $a \in \text{domain of } f(x)$, if

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

i.e., LHL = RHL = value of a function at $x = a$

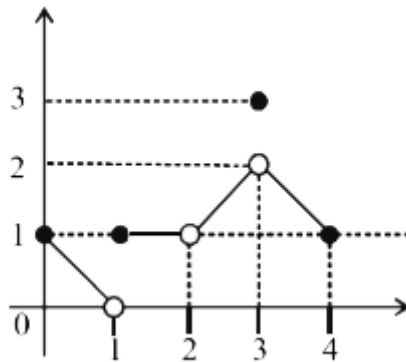
or
$$\lim_{x \rightarrow a} f(x) = f(a)$$

1.1 Reasons of discontinuity

If $f(x)$ is not continuous at $x = a$, we say that $f(x)$ is discontinuous at $x = a$

There are following possibilities of discontinuity:

1. $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ exist but they are not equal.
2. $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ exists and are equal but not equal to $f(a)$
3. $f(a)$ is not defined.
4. At least one of the limits does not exist. The graph of the function will show a break at the location of discontinuity from a geometric standpoint.



The graph as shown is discontinuous at $x=1, 2$ and 3 .

2. PROPERTIES OF CONTINUOUS FUNCTIONS

Let $f(x)$ and $g(x)$ be continuous functions at $x = a$. Then,

1. $cf(x)$ is continuous at $x = a$, where c is any constant.
2. $f(x) \pm g(x)$ is continuous at $x = a$.
3. $f(x) \cdot g(x)$ is continuous at $x = a$.
4. $f(x)/g(x)$ is continuous at $x = a$, provided $g(a) \neq 0$.-
5. Assuming $f(x)$ be continuous on $[a, b]$ in such a way that the function $f(a)$ and $f(b)$ will be at opposite signs, then there will exist at least one solution of equation $f(x) = 0$ in the open interval (a, b)

4. CONTINUITY IN AN INTERVAL

(a) A function f is said to be continuous in (a, b) if f is continuous at each and every point $\in (a, b)$

(b) A function f is said to be continuous in a closed interval $[a, b]$ if :

(1) f is continuous in the open interval (a, b) and

(2) f is right continuous at 'a' i.e. $\lim_{x \rightarrow a^+} f(x) = f(a) = a$ finite quantity

(3) f is left continuous at 'b'; i.e. $\lim_{x \rightarrow b^-} f(x) = f(b) = a$ finite quantity

5. A LIST OF CONTINUOUS FUNCTIONS

Function $f(x)$	Interval in which $f(x)$ is continuous
1. constant c	$(-\infty, \infty)$
2. x^n , n is an integer ≥ 0	$(-\infty, \infty)$
3. x^{-n} , n is a positive integer	$(-\infty, \infty) - \{0\}$
4. $ x - a $	$(-\infty, \infty)$
5. $P(x) = a_0x^n + a_1x^{n-1} + \dots + a_n$	$(-\infty, \infty)$
6. $\frac{p(x)}{q(x)}$ where $p(x)$ and $q(x)$ are polynomial in x and $q(x) \neq 0$	$(-\infty, \infty) - \{x; q(x) = 0\}$
7. $\sin x$	$(-\infty, \infty)$
8. $\cos x$	$(-\infty, \infty)$

$$9. \tan x$$

$$(-\infty, \infty) - \left\{ (2n+1)\frac{\pi}{2} : n \in \mathbb{Z} \right\}$$

$$10. \cot x$$

$$(-\infty, \infty) - \{n\pi : n \in \mathbb{Z}\}$$

$$11. \sec x$$

$$(-\infty, \infty) - \{(2n+1)\pi/2 : n \in \mathbb{Z}\}$$

$$12. \operatorname{cosec} x$$

$$\pi/2 : n \in \mathbb{Z}$$

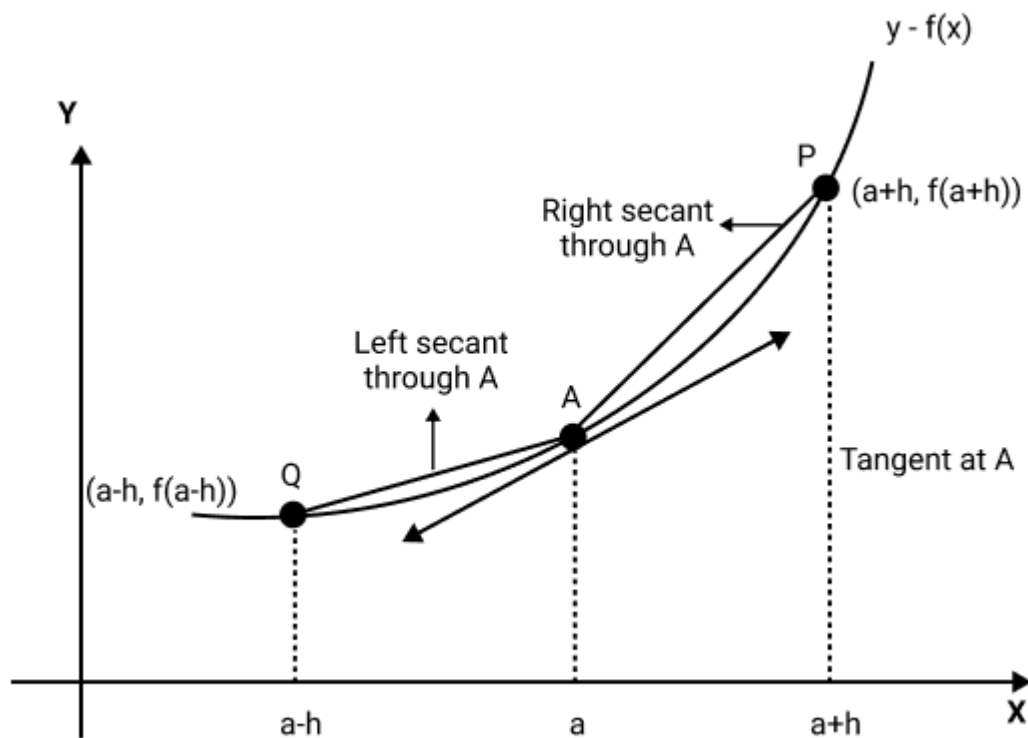
$$13. e^x$$

$$(-\infty, \infty) - \{n\pi : n \in \mathbb{Z}\}$$

$$14. \log_c x$$

$$(-\infty, \infty) \cup (0, \infty)$$

$$-x$$



- Slope of Right hand secant $= \frac{f(a+h) - f(a)}{h}$ as $h \rightarrow 0, P \rightarrow A$ and secant (AP) $\rightarrow$ tangent at A

$$\Rightarrow \text{Right hand derivative} = \lim_{h \rightarrow 0} \left(\frac{f(a+h) - f(a)}{h} \right)$$

= Slope of tangent at A (when approached from right) $f'(a^+)$

- Slope of Left hand secant $= \frac{f(a-h) - f(a)}{-h}$ as $h \rightarrow 0, Q \rightarrow A$ and secant AQ $\rightarrow$ tangent at A

$$\Rightarrow \text{Left hand derivative} = \lim_{h \rightarrow 0} \left(\frac{f(a-h) - f(a)}{-h} \right)$$

= Slope of tangent at A (when approached from left) $f'(a^-)$

Thus, $f(x)$ is differentiable at $x = c$.

$$\Leftrightarrow \lim_{x \rightarrow c} \frac{f(x) - f(c)}{(x - c)} \text{ exists finitely}$$

$$\Leftrightarrow \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{(x - c)} = \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{(x - c)}$$

$$\Leftrightarrow \lim_{h \rightarrow 0} \frac{f(c - h) - f(c)}{-h} = \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h}$$

Hence, $\lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{(x - c)} = \lim_{h \rightarrow 0} \frac{f(c - h) - f(c)}{-h}$ is called the left hand derivative of $f(x)$

at $x = c$ and is denoted by $f'(c^-)$ or $Lf'(c)$ While, $\lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{(x - c)} = \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h}$ is called the right hand derivative of $f(x)$ at $x = c$ and is denoted by $f'(c^+)$ or $Rf'(c)$

If $f'(c^-) \neq f'(c^+)$, we say that $f(x)$ is not differentiable at $x = c$.

2. DIFFERENTIABILITY IN A SET

1. A function $f(x)$ defined on an open interval (a, b) is said to be differentiable or derivable in open interval (a, b) , if it is differentiable at each point of (a, b)

2. A function $f(x)$ defined on closed interval $[a, b]$ is said to be differentiable or derivable. "If f is derivable in the open interval (a, b) and also the end points a and b , then f is said to be derivable in the closed interval $[a, b]$ "

i.e., $\lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{(x - a)}$ and $\lim_{x \rightarrow b^-} \frac{f(x) - f(b)}{(x - b)}$, both exist.

A function f is said to be a differentiable function if it is differentiable at every point of its domain.

Note:

1. If $f(x)$ and $g(x)$ are derivable at $x = a$ then the functions $f(x) + g(x)$, $f(x) - g(x)$, $f(x) \cdot g(x)$ will also be derivable at $x = a$ and if $g(a) \neq 0$ then the function $f(x)/g(x)$ will also be derivable at $x = a$

2. If $f(x)$ is differentiable at $x = a$ and $g(x)$ is not differentiable at $x = a$, then the product function $F(x) = f(x) \cdot g(x)$ can still be differentiable at $x = a$. E.g. $f(x) = x$ and $g(x) = |x|$

DIFFERENTIATION:

1. DEFINITION

(a) Let us consider a function $y = f(x)$ defined in a certain interval. It has a definite value for each value of the independent variable x in this interval.

Now, the ratio of the function's increment to the independent variable's increment,

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Now, as $\Delta x \rightarrow 0, \Delta y \rightarrow 0$ and $\frac{\Delta y}{\Delta x} \rightarrow$ finite quantity, then derivative $f'(x)$ exists and is

denoted by y' or $f'(x)$ or $\frac{dy}{dx}$ Thus, $f'(x) = \lim_{x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ (if it exists)

for the limit to exist,

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{-h}$$

(Right Hand derivative) (Left Hand derivative)

(b) The derivative of a given function f at a point $x = a$ of its domain is defined as:

$$\text{Limit}_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}, \text{ provided the limit exists is denoted by } f'(a)$$

Note that alternatively, we can define

$$f'(a) = \text{Limit}_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}, \text{ provided the limit exists.}$$

This method is called first principle of finding the derivative of $f(x)$

2. DERIVATIVE OF STANDARD FUNCTION

$$(i) \frac{d}{dx}(x^n) = n \cdot x^{n-1}; x \in \mathbb{R}, n \in \mathbb{R}, x > 0$$

$$(ii) \frac{d}{dx}(e^x) = e^x$$

$$(iii) \frac{d}{dx}(a^x) = a^x \cdot \ln a (a > 0)$$

$$(iv) \frac{d}{dx}(\ln |x|) = \frac{1}{x}$$

$$(v) \frac{d}{dx}(\log_a |x|) = \frac{1}{x} \log_a e$$

$$(vi) \frac{d}{dx}(\sin x) = \cos x$$

$$(vii) \frac{d}{dx}(\cos x) = -\sin x$$

$$(viii) \frac{d}{dx}(\tan x) = \sec^2 x$$

$$(ix) \frac{d}{dx}(\sec x) = \sec x \cdot \tan x$$

$$(x) \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$$

$$(xi) \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$(xii) \frac{d}{dx}(\text{constant}) = 0$$

$$(xiii) \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1$$

$$(xiv) \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, \quad -1 < x < 1$$

$$(xv) \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}, \quad x \in \mathbb{R}$$

$$(xvi) \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}, \quad x \in \mathbb{R}$$

$$(xvii) \frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1$$

$$(xviii) \frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}, \quad |x| > 1$$

(xix) Results:

If the inverse functions $f(g)$ are defined by $y = f(x); x = g(y)$. Then

$$g(f(x)) = x \Rightarrow g'(f(x)) \cdot f'(x) = 1$$

This result can also be written as, if $\frac{dy}{dx}$ exists and $\frac{dy}{dx} \neq 0$, then $\frac{dx}{dy} = 1 / \left(\frac{dy}{dx} \right)$ or

$$\frac{dy}{dx} \cdot \frac{dx}{dy} = 1 \text{ or } \frac{dy}{dx} = 1 / \left(\frac{dx}{dy} \right) \left[\frac{dx}{dy} \neq 0 \right]$$

3. THEOREMS ON DERIVATIVES

If u and v are derivable functions of x , then,

(i) Term by term differentiation : $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$

(ii) Multiplication by a constant $\frac{d}{dx}(Ku) = K \frac{du}{dx}$, where K is any constant

(iii) "Product Rule" $\frac{d}{dx}(u.v) = u \frac{dv}{dx} + v \frac{du}{dx}$ known as In general,

(a) If $u_1, u_2, u_3, u_4, \dots, u_n$ are the functions of x , then

$$\begin{aligned} & \frac{d}{dx}(u_1 \cdot u_2 \cdot u_3 \cdot u_4 \dots u_n) \\ &= \left(\frac{du_1}{dx}\right)(u_2 u_3 u_4 \dots u_n) + \left(\frac{du_2}{dx}\right)(u_1 u_3 u_4 \dots u_n) \\ &+ \left(\frac{du_3}{dx}\right)(u_1 u_2 u_4 \dots u_n) + \left(\frac{du_4}{dx}\right)(u_1 u_2 u_3 u_5 \dots u_n) \\ &+ \dots + \left(\frac{du_n}{dx}\right)(u_1 u_2 u_3 \dots u_{n-1}) \end{aligned}$$

(iv) Quotient Rule

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\left(\frac{du}{dx}\right) - u\left(\frac{dv}{dx}\right)}{v^2} \text{ where } v \neq 0 \text{ known as}$$

(b) Chain Rule : If $y = f(u), u = g(w), w = h(x)$ then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dw} \cdot \frac{dw}{dx}$

or $\frac{dy}{dx} = f'(u) \cdot g'(\quad) \cdot h'(x)$

Note:

In general if $y = f(u)$ then $\frac{dy}{dx} = f'(u) \cdot \frac{du}{dx}$

4. METHODS OF DIFFERENTIATION

4.1 Derivative by using Trigonometrical Substitution

The use of trigonometrical transforms before differentiation greatly reduces the amount of labour required. The following are some of the most significant findings:

$$(i) \sin 2x = 2 \sin x \cos x = \frac{2 \tan x}{1 + \tan^2 x}$$

$$(ii) \cos 2x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

$$(iii) \tan 2x = \frac{2 \tan x}{1 - \tan^2 x}, \tan^2 x = \frac{1 - \cos 2x}{1 + \cos 2x}$$

$$(iv) \sin 3x = 3 \sin x - 4 \sin^3 x$$

$$(v) \cos 3x = 4 \cos^3 x - 3 \cos x$$

$$(vi) \tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$$

$$(vii) \tan\left(\frac{\pi}{4} + x\right) = \frac{1 + \tan x}{1 - \tan x}$$

$$(viii) \tan\left(\frac{\pi}{4} - x\right) = \frac{1 - \tan x}{1 + \tan x}$$

$$(ix) \sqrt{(1 \pm \sin x)} = \left| \cos \frac{x}{2} \pm \sin \frac{x}{2} \right|$$

$$(x) \tan^{-1} x \pm \tan^{-1} y = \tan^{-1} \left(\frac{x \pm y}{1 \mp xy} \right)$$

$$(xi) \sin^{-1} x \pm \sin^{-1} y = \sin^{-1} \left\{ x \sqrt{1 - y^2} \pm y \sqrt{1 - x^2} \right\}$$

$$(xii) \cos^{-1} x \pm \cos^{-1} y = \cos^{-1} \left\{ xy \mp \sqrt{1 - x^2} \sqrt{1 - y^2} \right\}$$

$$(xiii) \sin^{-1} x + \cos^{-1} x = \tan^{-1} x + \cot^{-1} x = \sec^{-1} x + \operatorname{cosec}^{-1} x = \pi / 2$$

$$(xiv) \sin^{-1} x = \operatorname{cosec}^{-1}(1/x); \cos^{-1} x = \sec^{-1}(1/x); \tan^{-1} x = \cot^{-1}(1/x)$$

Note:

Some standard substitutions:

Expressions	Substitutions
-------------	---------------

$\sqrt{(a^2 - x^2)}$	$x = a \sin \theta$ or $a \cos \theta$
----------------------	--

$\sqrt{(a^2 + x^2)}$	$x = a \tan \theta$ or $a \cot \theta$
----------------------	--

$\sqrt{(x^2 - a^2)}$	$x = a \sec \theta$ or $a \operatorname{cosec} \theta$
----------------------	--

$\sqrt{\left(\frac{a+x}{a-x}\right)}$ or $\sqrt{\left(\frac{a-x}{a+x}\right)}$	$x = a \cos \theta$ or $a \cos 2\theta$
--	---

$\sqrt{(a-x)(x-b)}$	or $x = a \cos^2 \theta + b \sin^2 \theta$
---------------------	--

$\sqrt{\left(\frac{a-x}{x-b}\right)}$ or $\sqrt{\left(\frac{x-a}{a-x}\right)}$	
--	--

$\sqrt{(x-a)(x-b)}$	or $x = a \sec^2 \theta - b \tan^2 \theta$
---------------------	--

$\sqrt{\left(\frac{x-a}{x-b}\right)}$ or $\sqrt{\left(\frac{x-a}{x-a}\right)}$	
--	--

$\sqrt{(2ax - x^2)}$	$x = a(1 - \cos \theta)$
----------------------	--------------------------

4.2 Logarithmic Differentiation

To find the derivative of:

If $y = \{f_1(x)\}^{f_2(x)}$ or $y = f_1(x) \cdot f_2(x) \cdot f_3(x) \dots$

or $y = \frac{f_1(x) \cdot f_2(x) \cdot f_3(x) \dots}{g_1(x) \cdot g_2(x) \cdot g_3(x) \dots}$ then it's easier to take the function's logarithm first and

then differentiate. This is referred to as the logarithmic function's derivative.

Ch – 5 Continuity and Differentiability

Continuity and differentiability, chain rule, derivative of inverse trigonometric functions, like $\sin^{-1}x$, $\cos^{-1}x$, $\tan^{-1}x$, derivative of implicit functions. Concept of exponential and logarithmic functions. Derivatives of logarithmic and exponential functions. Logarithmic differentiation, derivative of functions expressed in parametric forms. Second order derivatives.

Formulae for Limits

$$\begin{array}{llll} \text{(a)} \lim_{x \rightarrow 0} \cos x = 1, & \text{(b)} \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, & \text{(c)} \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1, & \text{(d)} \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = 1 \\ \text{(e)} \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 1 & \text{(f)} \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a, a > 0 & \text{(g)} \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 & \text{(h)} \lim_{x \rightarrow 0} \frac{\log_e e^{(1+x)}}{x} = 1 \\ \text{(i)} \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \end{array}$$

Important terms and facts about Limits & Continuity of a function

- (i) For a function $f(x)$, $\lim_{x \rightarrow m} f(x)$ exists if $\lim_{x \rightarrow m^-} f(x) = \lim_{x \rightarrow m^+} f(x)$
(ii) A function $f(x)$ is continuous at a point $x = m$ if,

$$\lim_{x \rightarrow m^-} f(x) = \lim_{x \rightarrow m^+} f(x) = f(m)$$

Where $\lim_{x \rightarrow m^-} f(x)$ or $\lim_{h \rightarrow 0} f(m - h)$ is **Left Hand Limit** of $f(x)$ at $x = m$ and

$\lim_{x \rightarrow m^+} f(x)$ or $\lim_{h \rightarrow 0} f(m + h)$ is **Right Hand Limit** of $f(x)$ at $x = m$ ($0 < h < \infty$)

Also $f(m)$ is the value of the function $f(x)$ at $x = m$.

Formulae for Derivatives

$$\begin{array}{lll} \text{(a)} \frac{d}{dx} (x^n) = nx^{n-1} & \text{(b)} \frac{d}{dx} (k) = 0, \text{ where } k \text{ is any constant.} & \text{(e)} \frac{d}{dx} (e^x) = e^x \\ \text{(d)} \frac{d}{dx} (\log_e x) = \frac{1}{x} & \text{(e)} \frac{d}{dx} (\sin x) = \cos x & \text{(f)} \frac{d}{dx} (\cos x) = -\sin x \\ \text{(g)} \frac{d}{dx} (\tan x) = \sec^2 x & \text{(h)} \frac{d}{dx} (\sec x) = \sec x \tan x & \text{(i)} \frac{d}{dx} (\cot x) = -\operatorname{cosec}^2 x \\ \text{(j)} \frac{d}{dx} (\operatorname{cosec} x) = -\operatorname{cosec} x \cot x & \text{(k)} \frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} & \text{(l)} \frac{d}{dx} (\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}} \\ \text{(m)} \frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2} & \text{(n)} \frac{d}{dx} (\cot^{-1} x) = -\frac{1}{1+x^2} & \text{(o)} \frac{d}{dx} (\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}} \\ \text{(p)} \frac{d}{dx} (\operatorname{cosec}^{-1} x) = -\frac{1}{x\sqrt{x^2-1}} & \text{(q)} \frac{d}{dx} (\sqrt{x}) = \frac{1}{2\sqrt{x}} & \text{(r)} \frac{d}{dx} \left(\frac{1}{x}\right) = -\frac{1}{x^2} \end{array}$$